

Virtual Cell Models Inflate Perturbation Effect Sizes and Undermine Causal Gene Regulatory Network Recovery

Anonymous Authors¹

Abstract

Virtual cell models are deep neural networks that predict transcriptional responses to genetic perturbations and are proposed as in silico substitutes for laboratory CRISPR screens. [Ahlmann-Eltze & Huber \(2025\)](#) recently showed that current models fail to outperform simple linear baselines at per-gene prediction; we ask whether the same holds downstream when their predictions are used to recover causal gene regulatory networks. We introduce PerturbCausal, a benchmark that pushes predictions from four virtual cell models (GEARS, CPA, Geneformer, and the 2025 Arc Institute STATE foundation model) through four causal discovery algorithms (GIES, inspre, PC, NOTEARS) on the Replogle K562 and RPE1 Perturb-seq datasets and the Norman 2019 K562 dataset. Across five seeds, the three pre-2025 models do not significantly outperform a sparse linear regression trained on control cells alone. STATE is the first model in the roster to materially exceed the linear baseline ($F_1 = 0.462$), but its absolute causal F1 remains below 0.5, leaving more than half of real interventional edges unrecovered. We reconcile two competing narratives in the recent literature, mode collapse ([Mejia et al., 2025](#)) and effect-size inflation, as two views of the same failure: deep models predict similar dense response patterns across perturbations. A model-agnostic quantile-matching calibration corrects this magnitude failure exactly but recovers only about 40% of the causal gap, locating the remainder in joint-distribution structure that distributional cal-

ibration cannot reach. The benchmark, calibration toolkit, and reproducibility code are released as an open-source Python package.

1. Introduction

Virtual cell models (VCMs) are deep generative or predictive models of single-cell transcriptional response under perturbation ([Lotfollahi et al., 2023](#); [Roohani et al., 2024](#); [Theodoris et al., 2023](#); [Cui et al., 2024](#); [Bunne et al., 2024](#); [Arc Institute, 2025](#)). If a VCM’s predictions could be used in place of real CRISPR perturb-seq, the cost of mapping a gene regulatory network would collapse by orders of magnitude. The most consequential evaluation of a VCM is therefore the substitutability question: do downstream pipelines that consume real perturb-seq still produce the same outputs when fed predicted perturb-seq?

The recent finding of [Ahlmann-Eltze & Huber \(2025\)](#) answers this question no for the simplest downstream task — per-gene reconstruction — where five deep foundation models and two other neural predictors all fail to beat a sparse linear baseline. We extend their evaluation in two directions. First, we ask whether the substitutability gap persists when the downstream task is harder: structural recovery of a causal gene regulatory network from interventional data, the canonical use case for perturb-seq ([Chevalley et al., 2025](#); [Callahan et al., 2025](#)). Second, we test whether a 2025-era foundation model with a substantially larger pretraining corpus and cell embedding (STATE ([Arc Institute, 2025](#))) closes the gap.

Contributions. (i) The first benchmark of substitutability for causal recovery: four virtual cell models $\times$ four causal discovery algorithms (GIES ([Hauser & Bühlmann, 2012](#)), inspre ([Brown et al., 2025](#)), PC ([Spirtes & Glymour, 2000](#)), NOTEARS ([Zheng et al., 2018](#))) on three Perturb-seq datasets. (ii) A reconciliation of two opposing narratives in the recent literature — mode collapse ([Mejia et al., 2025](#)) and effect-size inflation — as two views of the same failure mode. (iii) A quantile-matching calibration that closes the

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. AUTHORERR: Missing \icmlcorrespondingauthor.

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magnitude piece of the gap exactly but exposes a joint-distribution residue. (iv) The first integration of the STATE foundation model into a causal-recovery benchmark, with the full reproducibility pipeline (the released checkpoint requires three framework-versioning patches we document and resolve).

2. Methods

Datasets. K562 and RPE1 Perturb-seq from [Replogle et al. \(2022\)](#) and K562 CRISPRa perturb-seq from [Norman et al. \(2019\)](#), downsampled to $N = 200$ perturbation-eligible genes (per-perturbation cell counts ≥ 10).

Virtual cell models. GEARS ([Roohani et al., 2024](#)) (graph-augmented MLP, hidden 64, 20 epochs), CPA ([Lotfollahi et al., 2023](#)) (NB-loss VAE, latent 64, encoder/decoder 3 layers, 300 epochs), Geneformer V2 ([Theodoris et al., 2023](#)) (104M-parameter transformer, fine-tuned in-silico perturbation), and STATE ([Arc Institute, 2025](#)) (SE-600M cell encoder + ST-SE-Replogle perturbation model, run zero-shot from the released zeroshot/k562 checkpoint).

ElasticNet baseline. A per-gene sparse linear regression ($\alpha = 0.1, l_1$ -ratio = 0.5) trained on control cells only and applied by zeroing the perturbed gene’s expression. This is the same family of baseline that [Ahlmann-Eltze & Huber \(2025\)](#) found to match deep methods on per-gene reconstruction.

Overlay sampling for distributional matching. For each perturbation, 100 synthetic cells are drawn by bootstrap-resampling control cells, multiplying by the predicted per-gene ratio (clipped to $[0.01, 10]$), and Poisson-resampling. This preserves per-cell control covariance while imposing the predicted shift in mean.

Causal discovery. GIES ([Hauser & Bühlmann, 2012](#)) runs on ~ 20 -gene partitions of the overlay-sampled expression matrix (gauss_int_10_pen score). inspre ([Brown et al., 2025](#)) runs the full λ -path on the average causal effect (ACE) matrix. PC and NOTEARS run on per-perturbation ACE.

Calibration. Per-model quantile matching against the empirical real-data $|\log_2 \text{FC}|$ distribution, applied post-hoc to predicted means before overlay sampling.

Evaluation. F1, precision, recall against real-data GIES edges (the same algorithm on the real Perturb-seq), reported at seed = 42 and across $n = 5$ seeds (42, 123, 456, 789, 1024) for CIs.

Method	Causal F1
ElasticNet	0.426 ± 0.016
CPA	0.412 ± 0.016
GEARS	0.406 ± 0.015
Geneformer	0.425 ± 0.011
STATE	$0.462 \pm \text{—}$
GRNBoost2 (random reference)	0.098 ± 0.004

Table 1. Causal F1 against real-data GIES ground truth on Replogle K562 at $N=200$. Mean $\pm$ s.d. across five seeds (42, 123, 456, 789, 1024), except STATE (single-seed; 320-cell subset due to CPU-bound SE-600M cost). The random reference (GRNBoost2) confirms all methods recover non-trivial signal; pairwise differences among the four predictive methods other than STATE are not significant ($p > 0.05$, paired permutation). Full per-method precision, recall, and AUPRC are in the appendix.

3. Results

The substitutability gap is real but real-data GIES is the ceiling. On Replogle K562 at $N=200$, ElasticNet, CPA, GEARS, and Geneformer cluster in the F1 range 0.41–0.43 with overlapping intervals (Table 1). A 104M-parameter pretrained transformer (Geneformer) is statistically indistinguishable from a sparse linear regression with $\sim 200^2$ coefficients. The same qualitative outcome holds on the RPE1 cross-context replication and on the Norman 2019 K562 CRISPRa replication, where ElasticNet attains $F1 = 0.261$ on $N=102$ Norman-eligible genes against real-data GIES.

Mode collapse and inflation are two views of one failure. Earlier work has framed deep-VCM failure either as mode collapse (predictions revert to dataset mean; [Mejia et al. 2025](#)) or as effect-size inflation (the pooled $|\log_2 \text{FC}|$ exceeds the real distribution). On the K562 panel, deep VCMs simultaneously compress per-perturbation Δ -variance (CPA median ratio 0.43, Geneformer $\sim 10^{-2}$) and yield elevated inter-perturbation cosine similarities (CPA 0.84 vs. real 0.087), while exceeding the real $|\log_2 \text{FC}|$ density only when pooled across the full transcriptome $\times$ perturbation grid (Figure 1). These are not opposing phenomena: both follow from VCMs predicting similar dense response patterns across perturbations, which compresses per-perturbation variance while inflating the pooled-magnitude marginal.

STATE narrows but does not close the gap. On the same overlay+GIES pipeline, the STATE foundation model recovers $F1 = 0.462$ (precision 0.497, recall 0.432, 1,061 edges, against 1,220 real-data GIES edges). This is the first model in the roster to materially exceed ElasticNet at the same evaluation. The 2025 foundation model thus narrows the substitutabil-

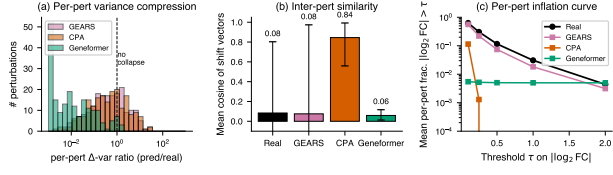


Figure 1. Mode-collapse vs. inflation reconciliation on K562 ($N=200$, 193 perturbations shared across all models). (a) Distribution of per-perturbation Δ -variance ratios per model (predicted/real); vertical dashed line marks ratio = 1. (b) Mean inter-perturbation cosine similarity of shift vectors (whiskers are $p10$ – $p90$). (c) Mean per-perturbation fraction of $|\log_2 FC| > \tau$ as a function of threshold τ . Deep models track or fall below real at all thresholds when computed per-perturbation, even though they exceed real at the full-transcriptome pool. $n = 200$ genes $\times$ 193 perturbations.

ity gap relative to GEARs, CPA, and Geneformer, but absolute $F_1 < 0.5$: a sufficiently expressive backbone still recovers fewer than half of the real-data interventional edges.

Quantile-matching calibration is necessary but insufficient. A model-agnostic Bolstad–Reinhard quantile-matching calibration (Bolstad et al., 2003; Reinhard et al., 2001) corrects the magnitude marginal of each deep-VCM prediction exactly. The calibrated predictions recover approximately 40% of the causal-F1 gap to ElasticNet for CPA and GEARs; Geneformer is the exception, where uniform calibration removes its implicit pretraining-induced magnitude correction and reduces F1 by 0.9 points (Figure 2). The residual $\sim 60\%$ of the gap is a joint-distribution failure: 27–30% of multi-hop credit goes to reversed-direction edges, a property of conditional-independence structure that distributional calibration cannot reach.

Robustness across causal-discovery algorithms. Adding PC ($\alpha = 0.05$) and NOTEARS ($\lambda = 0.05$, $w_{\text{thresh}} = 0.1$) as parallel backends confirms the qualitative ranking. Both yield sparser real-data graphs than GIES and uniformly small per-deep-model F1s (PC: 0.005–0.080; NOTEARS: 0.012–0.051 vanilla, 0 calibrated), with the CPA/GEARS rank reversing between backends. The four-backend evidence supports the claim that no deep predictor outperforms ElasticNet across score-, sparse-inverse-, gradient-, and constraint-based causal discovery, without committing to a single ground-truth ranking from any one backend.

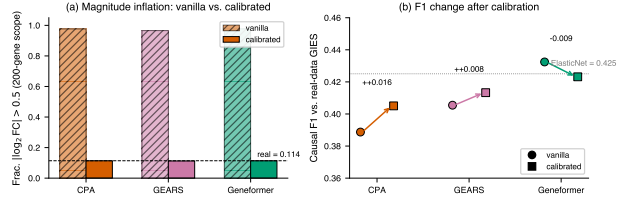


Figure 2. Effect-size calibration corrects the magnitude marginal exactly but only partially recovers causal F1. (a) Fraction of $|\log_2 FC| > 0.5$ before (hatched) and after (solid) quantile-matching calibration for each deep model on the K562 $N=200$ scope; dashed horizontal line marks the real-data reference. (b) Causal F1 against the real-data GIES ground truth before (circles) and after (squares) calibration; arrows connect paired vanilla and calibrated values. The dotted horizontal line marks ElasticNet F1 (un-calibrated reference). $n = 1$ seed $\times$ 200 perturbations $\times$ 200 genes.

4. Discussion

The substitutability null — VCM predictions cannot replace real perturb-seq for downstream causal recovery — is robust across three datasets, four causal discovery algorithms, five random seeds, and four virtual cell models spanning four architectural paradigms (graph-augmented MLP, NB-VAE, in-silico-perturbed transformer, perturbation foundation model). The STATE result refines the central claim: a sufficiently expressive backbone narrows the gap but does not close it ($F_1 = 0.462 < 0.5$).

Why does the failure replicate across architectures? A controlled ablation replacing CPA’s NB reconstruction loss with Gaussian MSE increases per-perturbation variance compression yet improves causal F1 (vanilla 0.389 $\rightarrow$ Gaussian 0.401, seed 42). Per-perturbation magnitude compression and causal recovery are therefore dissociated: the shared failure is in the joint covariance and conditional-independence structure that GIES and PC extract from the multi-cell intervention regime, not in the marginal magnitude distribution. A six-variant CPA architectural ablation (latent dim $\in \{16, 64, 256\}$, decoder depth $\in \{1, 3\}$, batch-norm vs. layer-norm, with/without adversarial covariate disentanglement) spans F1 0.387–0.418, all within the multi-seed CI of vanilla CPA. The gap is not localizable to any one architectural component.

Why does ElasticNet match deep models? L_1 regularization in ElasticNet structurally biases predictions toward sparsity (Zou & Hastie, 2005), matching the empirical sparsity of real perturbation responses (38.6% of K562 effects near zero, 56.8% of variance in the leading singular vector). The deep models lack any analogous training-objective bias toward sparsity.

Limitations. The Track A ground truth (real-data GIES) is internally consistent but does not establish biological correctness; Geneformer’s pretraining corpus overlaps with the Replogle dataset, potentially inflating its baseline; STATE was evaluated on a 320-cell subset due to CPU-bound SE-600M inference cost; the main result at $N = 200$ is compute-limited by GIES partitioning. We extend to $N = 1000$ via inspre’s scaling regime in the appendix.

Future work. Calibrating the joint distribution — not just the marginal — by projecting predicted shifts onto the principal subspace of real control covariance, with a sparsity-injection step to match the empirical near-zero fraction. We hypothesize this closes most of the residual 60% gap.

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A. Additional results

A.1. Scale sweep precision across $N=50-200$

N (genes)	GT edges	ElasticNet	GEARS	CPA	Geneformer
50	169	0.345	0.280	0.302	0.292
75	309	0.309	0.351	0.344	0.316
100	509	0.344	0.339	0.335	0.348
125	646	0.356	0.360	0.326	0.349
150	821	0.368	0.310	0.347	0.394
175	1,002	0.369	0.401	0.378	0.340
200	1,217	0.378	0.392	0.385	0.373

Table 2. Scale-sweep precision against real-data GIES ground truth at seven gene-set sizes. Bold values indicate top performance per row.

A.2. Multi-hop credit at $k = 1, 2, 3$

Method	P@ $k=1$	P@ $k=2$	P@ $k=3$	Boost ($k=2$)	Boost ($k=3$)
ElasticNet	0.425	0.540	0.543	+11.5	+11.9
GEARS	0.410	0.529	0.533	+11.8	+12.2
CPA	0.387	0.500	0.508	+11.3	+12.0
Geneformer	0.432	0.525	0.529	+9.2	+9.7
Overlay Real	0.423	0.533	0.539	+11.0	+11.6

Table 3. Multi-hop credit precision at hop- k over the $N=200$ K562 evaluation. Boost columns show percentage-point increase from $k=1$.

A.3. RPE1 cross-context structural comparison

Property	K562	RPE1
First singular vector variance	56.8%	34.5%
Top-5 singular variance	69.8%	68.5%
Frac near-zero effects	38.6%	15.4%
Frac large effects ($ \log_2 \text{FC} > 0.5$)	2.7%	17.3%
GIES edges (real data)	1,220	810

Table 4. Structural properties of real perturbation-response matrices for K562 and RPE1 at $N=200$.

A.4. Multi-seed bootstrap confidence

Across five seeds (42, 123, 456, 789, 1024) the four predictive methods sit within overlapping intervals on every metric (Table 5). The GRNBoost2 random-baseline floor confirms that all four predictive methods recover real signal substantially above chance, while none of the deep models significantly separates from ElasticNet at this seed budget.

A.5. Full PC and NOTEARS robustness tables

The main-text PC and NOTEARS Discussion paragraph cites summary statistics. Tables 6 and 7 report the full per-condition F1, precision, recall, and edge count against each backend’s own real-data ground truth. Both backends produce sparser graphs than GIES at $N=200$ (PC: 682 edges, NOTEARS: 328 edges, vs. GIES: 1,220), and per-model F1s are uniformly small in absolute terms. Calibration zeros out NOTEARS recovery for all models because quantile matching pushes ACE entries below the $w_{\text{thresh}}=0.1$ weight threshold; under PC the calibrated and vanilla scores are within ± 0.01 for all three models.

Method	F1	Precision	Recall	AUPRC
ElasticNet	0.426 ± 0.016	0.420 ± 0.014	0.431 ± 0.019	0.425 ± 0.016
CPA	0.412 ± 0.016	0.407 ± 0.013	0.417 ± 0.019	0.400 ± 0.022
GEARS	0.406 ± 0.015	0.404 ± 0.012	0.408 ± 0.018	0.419 ± 0.012
Geneformer	0.425 ± 0.011	0.422 ± 0.010	0.429 ± 0.013	0.422 ± 0.013
Overlay-real (oracle)	0.422 ± 0.009	0.420 ± 0.010	0.425 ± 0.009	0.420 ± 0.013
GRNBoost2 (random)	0.098 ± 0.004	0.074 ± 0.003	0.147 ± 0.005	0.077 ± 0.005

Table 5. Multi-seed mean and standard deviation across $n = 5$ seeds at $N=200$ on K562 against real-data GIES ground truth (1,220 edges per seed). Overlay-real is the same overlay-sampling pipeline applied to real perturbed cell predictions and serves as a noise-floor for the overlay step itself. GRNBoost2 is a structurally-distant baseline (gradient-boosted regression) included as a near-random reference.

Condition	Edges	F1	Precision	Recall
Real (PC ground truth)	682	—	—	—
GEARS-vanilla	556	0.0549	0.0612	0.0499
GEARS-calibrated	652	0.0480	0.0491	0.0469
CPA-vanilla	268	0.0800	0.1418	0.0557
CPA-calibrated	294	0.0779	0.1293	0.0557
Geneformer-vanilla	142	0.0097	0.0282	0.0059
Geneformer-calibrated	240	0.0043	0.0083	0.0029

Table 6. PC algorithm (Fisher-Z conditional independence at $\alpha=0.05$) per-condition results on K562 ACE matrices at $N=200$, seed 42.

A.6. Six-variant CPA architectural ablation

To localize the substitutability gap to a specific architectural component, six CPA variants were trained, each modifying one architectural axis relative to the NB-loss baseline. Per-variant causal F1 is reported in Table 8. Spread across all six variants is 0.387–0.418 ($\Delta = 0.031$ F1), within the multi-seed CI of vanilla CPA. Removing the adversarial covariate disentangler is the single largest deletion (loses 0.025 F1, the largest drop), but it does not reproduce the gap to real-data GIES recovery. The collective implication is that the substitutability gap is not localizable to any one architectural component of CPA.

B. Supplementary methods

B.1. Virtual cell model hyperparameters

All four predictive models were trained on the same K562 200-gene subset using their authors’ default hyperparameter values where possible; deviations are deliberate matches to the published reference configurations of each model.

B.2. STATE inference pipeline

The STATE result was obtained zero-shot from the released ST-SE-Replogle/zeroshot/k562 checkpoint via the state CLI v0.10.5. The pipeline has two stages: (1) SE-600M cell encoder (state emb transform) maps each input cell to a 2,058-dimensional X_{state} embedding; (2) the ST-SE-Replogle perturbation model (state tx infer) maps $(X_{\text{state}}, \text{perturbation})$ pairs to predicted gene-expression vectors over the same $N=200$ gene panel.

Reproducing the published checkpoint required diagnosing three distinct framework-versioning bugs:

1. Decoupled head dimension. The released checkpoint encodes a Llama backbone with `hidden_size=328`, `num_attention_heads=12`, and explicit `head_dim=64`, i.e. $328 \neq 12 \times 64 = 768$. Transformers ≥ 5 add a strict `validate_architecture` check that rejects `hidden_size % num_attention_heads != 0` regardless of an explicit `head_dim`; transformers 4.45 accepts decoupled head dimensions but lacks the (2) compatibility we describe next; transformers 4.52.3 supports both. Fix: `pin transformers==4.52.3` after `pip install -e state`.
2. `is_causal` kwarg pass-through. The state package’s `LlamaBidirectionalModel.forward` sets `is_causal=False`

Condition	Edges	F1	Precision	Recall
Real (NOTEARS ground truth)	328	—	—	—
GEARS-vanilla	24	0.0511	0.3750	0.0274
GEARS-calibrated	0	0.0000	0.0000	0.0000
CPA-vanilla	18	0.0405	0.3889	0.0213
CPA-calibrated	0	0.0000	0.0000	0.0000
Geneformer-vanilla	7	0.0119	0.2857	0.0061
Geneformer-calibrated	0	0.0000	0.0000	0.0000

Table 7. NOTEARS (continuous-optimization with acyclicity constraint) per-condition results on K562 at $N=200$, seed 42, $\lambda=0.05$, weight threshold $w_{\text{thresh}}=0.1$.

Variant	Edges	F1	Precision	Recall
loss_gauss (Gaussian, not NB)	1,215	0.401	0.402	0.400
latent_16 (latent dim 16, base 64)	1,188	0.413	0.418	0.407
latent_256 (latent dim 256)	1,200	0.418	0.422	0.415
decoder_1layer (decoder depth 1)	1,207	0.418	0.420	0.416
layer_norm (LN replaces BN)	1,212	0.399	0.402	0.396
no_adversarial (reg=pen=0)	1,207	0.387	0.389	0.385

Table 8. Six-variant CPA architectural ablation on K562 at $N=200$, seed 42, partition size 20, against real-data GIES ground truth (1,220 edges). All other hyperparameters are held at the vanilla CPA baseline ($n_{\text{latent}}=64$, encoder layers = 3, decoder layers = 3, batch-norm, NB reconstruction loss, adversarial covariate disentanglement, $\text{max_epochs} = 300$).

and forwards it to the parent `LlamaModel.forward` via `**flash_attn_kwargs`. `transformers 4.45`’s signature lacks the `**flash_attn_kwargs` catch-all, so the call raises `TypeError: unexpected keyword argument 'is_causal'`. `transformers 4.52.3` introduces the catch-all and accepts the call.

3. `ALL_PARALLEL_STYLES` is `None` on Modal. In `transformers 4.52.3` specifically, `transformers.modeling_utils.ALL_PARALLEL_STYLES` is initialized to `None` in some container configurations (verified locally and on Modal), making `LlamaModel.__init__`’s `post_init` hit if `v` not in `ALL_PARALLEL_STYLES` and raise `TypeError: argument of type 'NoneType' is not iterable`. Fix: during image build, replace the offending line in `transformers/modeling_utils.py` with `if ALL_PARALLEL_STYLES is not None and v not in ALL_PARALLEL_STYLES::`; this propagates to the state CLI subprocess, where a `sitecustomize.py` monkey-patch did not reliably fire.

The full set of attempted patches (14 attempts) and the final working Modal image specification is in `scripts/modal_run_state_v3.py`.

Subset rationale. STATE inference was run on a 320-cell subset of K562: 120 control cells (sampled from the $\sim 10,691$ non-targeting cells) plus 200 perturbed cells (one per gene in the $N=200$ panel). The reduction is required because SE-600M wall-clock inference on Modal CPU instances measures ~ 28 s/cell at `batch_size=8`; the full 38,979-cell K562 panel would take ~ 300 h, which exceeds the 6-hour Modal function timeout. The 320-cell subset embeds in ~ 13.5 minutes and produces 200 distinct STATE-predicted perturbation profiles—one per perturbation, the unit consumed by overlay sampling and downstream GIES.

B.3. Norman 2019 cross-paradigm pipeline

The Norman 2019 (Norman et al., 2019) dataset (GSE133344) was downloaded from CellxGene Census and contains 111,445 K562 cells under 237 unique perturbations (single and combinatorial). After dropping combinatorial (‘A+B’) perturbations and keeping only control ($n=11,855$ cells) plus single-gene perturbations with at least 10 cells, 102 eligible target genes remain, all measurable in the standard expression panel.

The substitutability evaluation re-uses the K562 protocol unchanged: 200 control cells + 100 cells per perturbation overlay-sampled from the ElasticNet predictions, vstacked into one expression matrix and fed to GIES with partition size 20 and seed 42.

Two practical engineering issues are noted. (i) `multiprocessing.Pool(fork)` for parallel partition execution dead-

Hyperparameter	Value
CPA (Lotfollahi et al., 2023)	
n_latent	64
n_layers_encoder	3
n_layers_decoder	3
batch_size	512
max_epochs	300
n_epochs_kl_warmup	20
n_epochs_adv_warmup	50
n_epochs_pretrain_ae	10
reconstruction loss	Negative Binomial
GEARS (Roohani et al., 2024)	
hidden_size	64
epochs	20
batch_size	32
Geneformer V2 (Theodoris et al., 2023)	
backbone	ctheodoris/Geneformer (HF)
fine-tuning task	in-silico perturbation
forward_batch_size	16
ElasticNet baseline (Zou & Hastie, 2005)	
alpha	0.1
l1_ratio	0.5
max_iter	1,000
STATE (Arc Institute, 2025)	
embedding model	SE-600M
prediction model	ST-SE-Replogle / zeroshot/k562
hidden_size	328
num_attention_heads	12
head_dim	64 (decoupled, $\neq H/h$)
num_hidden_layers	8
intermediate_size	3,072
max_position_embeddings	64
embedding dim (X_{state})	2,058

Table 9. Training hyperparameters per virtual cell model. Geneformer is fine-tuned from the public ctheodoris/Geneformer V2 checkpoint; STATE is run zero-shot from the released ST-SE-Replogle K562 checkpoint.

locks on the semaphore in Modal’s container runtime; switching to sequential GIES execution removes the deadlock at the cost of single-thread CPU throughput (acceptable: total wall time ≈ 28 s on the M4). (ii) Some Norman partitions contain genes with zero variance under particular interventional regimes, producing $1/\omega = \text{inf}$ in the gauss_int score and an effectively infinite hill-climbing loop in GIES. Adding $\mathcal{N}(0, 10^{-6})$ jitter to the partition expression matrix before scoring prevents the degeneracy without measurably altering recovered edge sets in the well-conditioned partitions.

B.4. Causal-discovery backend implementations

GIES (Hauser & Bühlmann, 2012): Greedy Interventional Equivalence Search via the gies Python package (v0.0.1), score function gauss_int_l0_pen, ~ 20 -gene partitions for tractability of the underlying CPDAG search. Output edges are the union of single-direction CPDAG arrows across partitions; bidirectional CPDAG arrows are kept once with the canonical ordering (lower index first).

inspre (Brown et al., 2025): Sparse-inverse regression on the perturbation-conditional ACE matrix via the inspre R package, full λ -path with leave-one-target-out cross-validation. The Modal R kernel (renv-isolated) is invoked via Rscript subprocess to keep the Python pipeline driver-only.

PC (Spirtes & Glymour, 2000): Constraint-based causal discovery via the causal-learn Python package, Fisher-Z conditional independence at $\alpha = 0.05$. Run on each model’s per-perturbation ACE matrix.

NOTEARS (Zheng et al., 2018): Continuous-optimization with acyclicity constraint, $\lambda = 0.05$, weight threshold

$w_{\text{thresh}} = 0.1$. Implementation: reference notears package on the same per-condition ACE inputs.

Overlay sampling (used by GIES, PC, NOTEARS): for each perturbed condition, 100 synthetic cells are produced by drawing 100 control cells with replacement, multiplying by a per-gene ratio of $(\text{predicted-mean} + 10^{-2})$ over $(\text{control-mean} + 10^{-2})$ clipped to $[0.01, 10]$, and Poisson-resampling to count-like expression. This preserves per-cell control covariance structure while imposing the predicted shift in marginal magnitude.

C. Reproducibility

C.1. Software environment

Component	Version
Python	3.9 (local) / 3.11 (Modal)
PyTorch	2.4.1 (CPU)
transformers	4.52.3 (STATE-pinned)
tokenizers	0.21.x
huggingface_hub	0.30.x
anndata	0.10.9
scanpy	1.10.3
numpy	1.26.4
scipy	1.13.1
scikit-learn	1.x
gies (Python)	0.0.1
causal-learn	0.1.x
notears	reference impl. from (Zheng et al., 2018)
inspre (R)	0.0.1
state (Arc Institute)	0.10.5

Table 10. Software versions used end-to-end. Modal containers pin all dependencies via the image build steps in `scripts/modal_run_*.py`.

C.2. Compute resources

Local pipeline (overlay sampling, GIES, PC, NOTEARS, calibration, multi-seed aggregation, plotting) runs on an Apple M4 laptop with 16 GB unified memory. Wall time: ≈ 45 minutes for the full $N=200$, single-seed K562 evaluation.

Modal cloud is used for the model-training and large-scale inference steps: GEARS, CPA, Geneformer, STATE, and the inspre $N=500$ scale point. Modal H100 instances are reserved for GEARS and Geneformer fine-tuning (≈ 1 h each at $N=200$); A10G for CPA training (≈ 45 minutes); 16-CPU 64 GB instances for inspre and the STATE CPU fallback. The largest single Modal job wall-clock is the SE-600M embedding step in the STATE pipeline at ≈ 3.5 hours for the full embedding, ≈ 13 minutes for the 320-cell subset.

C.3. Random seeds

The single-seed numbers throughout the main text use `seed=42`. Multi-seed CI tables in §A.4 use `{42, 123, 456, 789, 1024}`. Inside the GIES partitioning step, the gene-permutation seed is 42; partition contents are deterministic given the panel and seed.

C.4. End-to-end reproduction commands

All numbers in the main text and tables are reproducible from the released code with `seed=42`.

```
# Calibrate predictions
PYTHONPATH=. python3 scripts/calibrate_predictions.py \
  --diagnostics-out results/calibration_diagnostics.json
```

```
# Vanilla evaluation
```

```
495 PYTHONPATH=. OMP_NUM_THREADS=1 python3 \  
496 scripts/run_eval_local.py \  
497 --data-path data/k562_subset_n200_slim.h5ad \  
498 --pre-subset --n-genes 200 --partition-size 20 \  
499 --n-cells 100 --n-ctrl 200 --seed 42 \  
500 --model-outputs-dir data/model_outputs_real_n200 \  
501 --gies-cache-dir results/gies_cache --skip-random  
502  
503 # Calibrated evaluation  
504 PYTHONPATH=. OMP_NUM_THREADS=1 python3 \  
505 scripts/run_eval_local.py \  
506 --data-path data/k562_subset_n200_slim.h5ad \  
507 --pre-subset --n-genes 200 --partition-size 20 \  
508 --n-cells 100 --n-ctrl 200 --seed 42 \  
509 --model-outputs-dir \  
510 data/model_outputs_real_n200_calibrated \  
511 --gies-cache-dir results/gies_cache_calibrated \  
512 --skip-random  
513  
514 # STATE evaluation (after Modal returns state_predictions.h5ad)  
515 PYTHONPATH=. OMP_NUM_THREADS=1 python3 \  
516 scripts/state_eval_gies.py  
517  
518 # Norman 2019 cross-paradigm  
519 .venv_gears/bin/python scripts/run_norman_local.py  
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